

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Final Examination of Semester 1

Year: 2019

Subject: 394173 Mathematics III

Section: 15-18

Date: 9 October 2019

Time: 13.00-16.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. There will be 9 pages (including this page) and they are worth 80 points.

- Answer the questions on this test papers.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators cannot be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

2. Consider the following infinite sequences, which may be convergent or divergent sequences. If it is a convergent sequence, find the limit of the sequence.

2.1 The sequence $\{a_n\}_{n=1}^{\infty}$ where $a_n = (-1)^{n+1} \left(\frac{9n^5 + 3n}{3n^5 + 7} - \frac{6n^2 - 1}{8n^3 + 5n^4} \right)$. (5 marks)

2.2 The sequence $\{b_n\}_{n=1}^{\infty}$ where $b_n = \sqrt{9^n - 3^n} + 1 - 3^n$. (5 marks)

4. 4.1 Let $a_1 + a_2 + a_3 + \dots + a_n + \dots$ be an arithmetic series.

If $a_1 + a_2 + a_3 + \dots + a_{10} = 100$ and $a_{11} + a_{12} + a_{13} + \dots + a_{20} = 300$,

find the value of a_{13} . (5 marks)

4.2 Let $a_1 + a_2 + a_3 + \dots + a_n$ be a geometric series with $a_2 = \frac{3}{2}$ and $S_3 = \frac{21}{4}$.

Find the summation of the first 4 terms ($r > 1$) of this series. (5 marks)

6. Given $\vec{u} = 3\vec{i} - \vec{j} + 2\vec{k}$, $\vec{v} = 2\vec{i} - 4\vec{j} + 3\vec{k}$ and $\vec{w} = -2\vec{i} - (2 - 3\lambda)\vec{j} + \lambda\vec{k}$.

6.1 Verify $\vec{u} \cdot \vec{v}$ and $\vec{u} \times \vec{v}$. (5 marks)

6.1 Find the value of λ such that $\vec{w} \cdot (\vec{u} \times \vec{v}) = 0$. (5 marks)

8. 8.1 Given the vector $\vec{a} = \vec{i} + \vec{j} - 2\vec{k}$, there is a vector $\vec{b}$ such that the angle between the vector $\vec{a}$ and the vector $\vec{b}$ is 60° and $(\vec{b} + \vec{a}) \cdot (\vec{b} - \vec{a}) = 10$. Find the value of $|\vec{b} \times \vec{a}|$. (5 marks)

8.2 Two vectors $\vec{u}$ and $\vec{v}$ satisfy $|\vec{u} + \vec{v}| = \sqrt{37}$ and $|\vec{u} - \vec{v}| = \sqrt{13}$. The angle between the vector $\vec{u}$ and the vector $\vec{v}$ is 60° . Determine the value of $|\vec{u}| - |\vec{v}|$. (5 marks)

ASSOC. PROF. DR. SANOE KOONPRASERT

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1. 1.1 Let $a_1, a_2, a_3, \dots, a_n$ be an arithmetic sequence. The summation of the first 5 terms is 20 and $a_4 - a_2 = 6$. Find the value of $a_5 + a_{10}$. (5 marks)

1.2 Assume that $k - 2, k - 6, 2k + 3$ is a geometric sequence, where $k > 0$ and $-6, x, y, z, 8$ is the arithmetic sequence with $a_w = 64$. Determine $(w - x - y - z)^k$. (5 marks)

2. Consider the following infinite sequences, which may be convergent or divergent sequences. If it is a convergent sequence, find the limit of the sequence.

2.1 The sequence $\{a_n\}_{n=1}^{\infty}$ where $a_n = (-1)^{n+1} \left(\frac{9n^5 + 3n}{3n^5 + 7} - \frac{6n^2 - 1}{8n^3 + 5n^4} \right)$. (5 marks)

2.2 The sequence $\{b_n\}_{n=1}^{\infty}$ where $b_n = \sqrt{9^n - 3^n + 1} - 3^n$. (5 marks)

4. 4.1 Let $a_1 + a_2 + a_3 + \dots + a_n + \dots$ be an arithmetic series.

If $a_1 + a_2 + a_3 + \dots + a_{10} = 100$ and $a_{11} + a_{12} + a_{13} + \dots + a_{20} = 300$,
find the value of a_{13} . (5 marks)

4.2 Let $a_1 + a_2 + a_3 + \dots + a_n$ be a geometric series with $a_2 = \frac{3}{2}$ and $S_3 = \frac{21}{4}$.

Find the summation of the first 4 terms ($r > 1$) of this series. (5 marks)

6. Given $\vec{u} = 3\vec{i} - \vec{j} + 2\vec{k}$, $\vec{v} = 2\vec{i} - 4\vec{j} + 3\vec{k}$ and $\vec{w} = -2\vec{i} - (2 - 3\lambda)\vec{j} + \lambda\vec{k}$.

6.1 Verify $\vec{u} \cdot \vec{v}$ and $\vec{u} \times \vec{v}$. (5 marks)

6.1 Find the value of λ such that $\vec{u} \cdot (\vec{u} \times \vec{v}) = 0$. (5 marks)

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